5.1.14 Are bitcoin transactions secure?

The transaction details, such as the sender and recipient addresses and the amount being transferred, are publicly visible on the *blockchain*, but the ownership of the ***bitcoin*** being transferred is verified through the use of cryptography.

What is Cryptography?

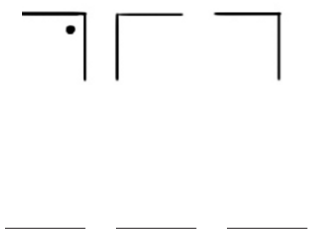
Cryptography is a way of keeping information secret by disguising it in code.

- **Encryption** is the process of taking information and putting it in a special code, making it unreadable to anyone without the correct decryption method. This is similar to locking a safe, where only the person with the correct key or combination can open it.
- On the other hand, **decryption** is the process of taking the coded information and **making it readable again**, like unlocking the safe and being able to read the information inside.

How to Solve
Pigpen Cipher

When solving the Pigpen Cipher, the player is given an encrypted message and a cipher. To decrypt the message, the player will find the symbol from the encrypted message on the cipher to find the decrypted letter.

- Example of an encrypted message:



For instance, let's say Arel and John want to keep a message hidden from someone named Ronny. They agree to use a secret key to disguise the message before sending it to each other. They could use a simple method like shifting each letter of the message down the alphabet, so that A becomes B, B becomes C, and so on. Only those with the key can decrypt the message, making it unreadable to Ronny. Though this method is not considered secure today, it illustrates the principle of private-key cryptography.